WORKFLOW I — NEW PORTFOLIO CREATION METHODOLOGY (STAGES 4.1 – 4.8: ONBOARDING, FUNNEL, AI & SLSQP MATH)

1. Investor Onboarding (Stage 4.1)

Capital & Horizon Mandate
V_total >= ₹10,000 • 3–360 Months

NLP Constraint Extractor
Parses OnTask & Sector Exclusions

5-Question Risk Score: S = sum(w_j * x_j) in [5..20]

Conservative	✓ Moderate	Aggressive
S in [5–9] (λ = 5.0)	S in [10–15] (λ = 2.5)	S in [16–20] (λ = 1.0)

Saved in DB-1 (user_profiles, risk_assessments)

Stage 4.1 Input -> Output
IN: Capital, Horizon, 5 Risk Answers & NLP Text
OUT: Structured Mandate & Policy Caps (DB-1)

2. Regime, Funnel & Factors (4.2–4.4)



VIX 14.2 (MOD)
Calm <13 | Mod 13–18
Vol 18–24 | Extr >24



Winsorized [5th, 95th] Sector Z-Scores in [-3, +3]

Quality Z_q (ROE) | Value Z_v (P/E, P/B) | Momentum Z_m (6M)

E[R_i] = R_base + 1.2(w_q Z_q + w_v Z_v + w_m Z_m) + 1.5(β_i - 1)

Stages 4.2–4.4 Input -> Output
IN: 2,639 Stocks (ADTV >= 25L), OHLCV & Ratios
OUT: 504 Liquid Pool, mu_i & Covariance Sigma_reg

3. AI Committee & SLSQP Math (4.5–4.7)

Nodes 1–8: Selects K=10..16 Stocks (Critic Rework <= 2)

Nodes 10–11: SciPy SLSQP & Whole Shares
- Max Sharpe -(w^T mu - R_f)/sigma_p | Min Var
- Integer Lots: Q_i = floor((w_i* V_total) / P_i)
- Greedy Pass: argmax(mu_i / P_i) for Cash

✓ Stock <= 25% • Sector <= 40% • Cash >= ₹0

Stages 4.5–4.7 Input -> Output
IN: 30–60 Shortlist, mu_i, Sigma_reg & Capital
OUT: Validated Integer Share Lots Q_i & Cash >= ₹0

4. Portfolio & 2-Level Why (Stage 4.8)



Return: 10.7% | Vol: 9.0%
Sharpe: 0.41 | HHI: 898

HDFCBANK: 32 sh (22.0%)
₹5,070 • CORE

RELIANCE: 10 sh (18.2%)
₹45,440 • CORE

TCS: 11 sh (16.0%)
₹39,006 • GROWTH

2-Level Stock-Why Modal
L1 Summary • L2 NSE/BSE Links

Stage 4.8 Input -> Output
IN: Validated Whole-Share Lots & Ratio Evidence
OUT: Committed PortfolioVersion & Broker CSV

Active Portfolio Tracked Every 300s

WORKFLOW II — EXISTING DEMAT PORTFOLIO DIAGNOSIS (STAGE 4.9)

5. CAS PDF / CSV Ingestion & HHI Risk Audit (Stage 4.9)

CAS PDF
or Broker CSV

CDSL / NSDL
pdfplumber 0.11.5

3-Step Entity Resolver (All 2,639 NSE/BSE Stocks)

1. Symbol/ISIN | 2. pg_trgm (>0.4) | 3. Levenshtein

HHI Concentration: HHI = sum((w_i * 100)^2)

<1,500 Safe | 1,500–2,500 Mod | >2,500 High Risk

7-Section Diagnostic Report & AI Copilot (/diagnosis/{id}/chat)
Flags Single-Stock (>25%), Sector (>40%) & Tier 4 Microcap Risks -> Suggests Swaps

Stage 4.9 Input -> Output
IN: Uploaded CAS PDF / CSV or Raw Holdings Text
OUT: PortfolioDiagnosisResponse (HHI, Health Score & Remediation Trades)

Rebalance

WORKFLOW III — CONTINUOUS 300s DRIFT MONITORING, STATUTORY REBALANCING & ALERTS (STAGES 4.10 – 4.12)

6. Continuous 300s Drift Radar (Stage 4.10)

Every 300s (5m)
09:15–15:30 IST
Lock: 847291

Live Valuation & Exposure Drift Check

✓ Healthy Corridor: |dw_i| < 5.0% & |ds_k| < 8.0%

! Drift Breach: |dw_i| >= 5.0% Stock OR |ds_k| >= 8.0% Sector

Freshness Guard: LIVE / END_OF_DAY / DELAYED_15M

Branch A: Hold Portfolio
Status: ON_TRACK (Zero Churn)

Branch B: Trigger Rebalance
Opens Stage 4.11 Trade Engine

Stage 4.10 Input -> Output
IN: Active Holdings Q_i, Live Quotes P_i(t) & NSE Holiday Calendar
OUT: PortfolioPerformanceSnapshot & Drift Breach Flags (DB-5)

Breach

7. Before/After Rebalancing & SEBI Costs (4.11–4.12)

BEFORE (Drifted)
IT Sector: 34.5% (+6.5%)
FMCG: 11.0%
! Drift >= 5% / 8% Cap

SELL -4
BUY +8

AFTER (Target v+1)
IT Sector: 28.0%
FMCG: 17.5%
✓ Aligned & Cash >= ₹0

Statutory Cost Engine (STT 0.10% • NSE/BSE • Stamp 0.015% • 18% GST)
✓ Guarantees Post-Trade Cash >= ₹0 (Trims 1 BUY Share if Needed) • 120m Alert Cooldown

Stages 4.11–4.12 Input -> Output
IN: Drifted Holdings, Target Weights w_i* & Broker Tariff
OUT: RebalancePreviewResponse, In-App Alert & Version (v+1)